

Meeting Room Booking System – Simple API Documentation

Welcome! This documentation explains how the **Meeting Room Booking System API** works in super simple terms.

Think of this API like a **reception desk** at an office. You can ask the receptionist:

1. "What rooms do we have?" 📄
 2. "What meetings are booked today?" 📅
 3. "Can I book Room A from 10:00 to 11:00?" 📝
 4. "Can you cancel my meeting?" ✕
 5. "When is Room A free for a 45-minute meeting?" ⌚
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Server Base URL

All request paths start with this base link:

```
http://localhost:5000/api
```

1. Get All Rooms

What does it do?

Ask the server for a list of all meeting rooms and how many people each room can fit.

- **HTTP Method:** GET
- **Endpoint:** /rooms
- **Full URL:** http://localhost:5000/api/rooms
- **Request Body:** None

Example Response (Success 200 OK):

```
{
  "rooms": [
    {
      "id": "room_id_101",
      "name": "Conference Room A",
      "capacity": 10
    },
  ],
}
```

```
{
  "id": "room_id_102",
  "name": "Boardroom B",
  "capacity": 20
}
```



2. Get Bookings List

What does it do?

Fetch scheduled meetings. You can look at all meetings, or filter by a specific **date** or **room**.

- **HTTP Method:** GET
- **Endpoint:** /booking
- **Full URL:** http://localhost:5000/api/booking
- **Optional Query Parameters:**
 - **date** : Filter by date e.g. 2026-10-01
 - **roomId** : Filter by room ID e.g. room_id_101

Example Request URL:

```
http://localhost:5000/api/booking?date=2026-10-01&roomId=room_id_101
```

Example Response (Success 200 OK):

```
{
  "count": 1,
  "data": [
    {
      "id": "booking_999",
      "title": "Sprint Planning",
      "date": "2026-10-01",
      "startTime": "09:00",
      "endTime": "10:30",
      "roomId": "room_id_101",
      "room": {
        "id": "room_id_101",
        "name": "Conference Room A",
        "capacity": 10
      }
    }
  ]
}
```

```
]
}
```

+ 3. Create a New Booking

What does it do?

Schedules a new meeting.

Server Rules:

1. Meetings must be between **09:00** and **18:00**.
2. **Start time** must be earlier than **End time**.
3. **No Overlaps!** If someone else already booked that room during your chosen time, the server rejects it and tells you who has the room.

- **HTTP Method:** `POST`
- **Endpoint:** `/booking`
- **Full URL:** `http://localhost:5000/api/booking`
- **Headers:** `Content-Type: application/json`

Request Body Example:

```
{
  "title": "Design Sync",
  "roomId": "room_id_101",
  "date": "2026-10-01",
  "startTime": "11:00",
  "endTime": "12:00"
}
```

Response A: Success (200 OK)

```
{
  "conflict": false,
  "message": "booking created successfully"
}
```

Response B: Conflict Error (400 Bad Request)

If the room is already booked at that time:

```
{
  "conflict": true,
  "status": 400,
  "conflictWith": {
    "title": "Sprint Planning",
    "startTime": "09:00",
    "endTime": "10:30"
  },
  "messages": "Conflict detected with booking \"Sprint Planning\" (09:00 - 10:30)"
}
```



4. Cancel / Delete a Booking

What does it do?

Deletes an existing meeting using its unique booking ID.

- **HTTP Method:** DELETE
- **Endpoint:** /booking
- **Full URL:** http://localhost:5000/api/booking
- **Headers:** Content-Type: application/json

Request Body Example:

```
{
  "id": "booking_999"
}
```

Example Response (Success 200 OK):

```
{
  "message": "booking deleted successfully"
}
```



5. Find Earliest Free Slot

What does it do?

Calculates the earliest available time window in a room for your requested meeting duration (in minutes) during office hours (09:00 to 18:00).

- **HTTP Method:** POST (or GET)
- **Endpoint:** /booking/next-available
- **Full URL:** http://localhost:5000/api/booking/next-available
- **Headers:** Content-Type: application/json

Request Body Example:

```
{
  "roomId": "room_id_101",
  "date": "2026-10-01",
  "duration": 45
}
```

Response A: Free Slot Found (200 OK)

```
{
  "AvailbleSlot": {
    "startTime": "10:30",
    "endTime": "11:15"
  }
}
```

Response B: No Slot Available (404 Not Found)

If the room is fully booked for the requested duration:

```
{
  "message": "sorry we dont have booking opt for your duration"
}
```

Summary Table

Action	HTTP Method	Endpoint Path	What you pass
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Action	HTTP Method	Endpoint Path	What you pass
Get All Rooms	GET	/api/rooms	Nothing
Get Bookings	GET	/api/booking?date=YYYY-MM-DD&roomId=XYZ	Query params
Create Booking	POST	/api/booking	JSON Body (title , roomId , date , startTime , endTime)
Delete Booking	DELETE	/api/booking	JSON Body (id)
Find Free Slot	POST	/api/booking/next-available	JSON Body (roomId , date , duration)